

Target-Aligned Full Conformal Bayes under Continuous Label Shift

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Abstract

Standard Full Conformal Bayes relies on exchangeability between the source observations and the target point. Under continuous label shift, weighting by the oracle label-density ratio transports rank aggregation to the target label distribution and restores target-domain coverage. However, the posterior predictive that defines the ranks may still reflect source-domain geometry, which can reduce interval efficiency. We propose Target-Aligned Full Conformal Bayes (TA-FCB), which keeps the weighted rank aggregation and additionally tilts each candidate-augmented posterior predictive before the scores are computed. For Bayesian ridge regression, candidate augmentation is a rank-one posterior update, and under an exponential label tilt the predictive tilt is a variance-scaled mean shift. Consequently, the procedure admits a closed-form implementation without candidate-wise refitting. The unrestricted oracle construction has finite-sample target-domain coverage under permutation-symmetric scoring, and the experiments use a shared finite-grid implementation. On two molecular property benchmarks, weighting recovers the coverage lost to controlled label shift, and candidate-wise tilting shortens intervals relative to weighting alone at comparable coverage. Estimated-ratio variants are evaluated separately and are not covered by the finite-sample guarantee.

Keywords: Conformal Bayes, Full Conformal Prediction, Label Shift, Posterior Predictive Tilting, Bayesian Ridge Regression, Uncertainty Quantification, Molecular Property Prediction

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1. Introduction

Conformal Bayes (Fong and Holmes, 2021) combines two ideas. Bayesian posterior predictive densities give an uncertainty-aware score over candidate outputs, and full conformal rank calibration turns those scores into prediction sets with finite-sample coverage under exchangeability, without distributional assumptions on the data-generating process. Fong and Holmes (2021) instantiated this connection within the full-conformal rank procedure: the nonconformity score is the negative log posterior predictive density, and candidate augmentation goes through an add-one-in (AOI) importance-sampling identity that reweights existing posterior samples instead of retraining the model. Validity then rests on exchangeability between the observed data and the test pair, and the rank comparisons are formed relative to the source label marginal.

Full conformal ranking is *candidate-wise*. For each candidate label y at a test input h^* , the test score is the negative log posterior predictive density of y under the model augmented with the pair (h^*, y) . The score that ranks the test pair against the calibration points therefore depends on the candidate, not only on the calibration labels. The AOI identity makes this candidate dependence computationally tractable. It does *not*, however, correct for distributional mismatch. AOI reweights samples for posterior computation, not the source-to-target label-density ratio.

This paper studies Conformal Bayes when the target label marginal differs from the source label marginal, and separates two places where the label-density ratio can act. The *post-ranking* operation changes the measure used to aggregate rank indicators. Oracle label-density-ratio weights replace the uniform masses in the full conformal p -value, which provides target-domain coverage under weighted exchangeability. The *pre-ranking* operation acts on the candidate-augmented posterior predictive that defines the Bayesian nonconformity score, tilting its geometry before any rank comparison is formed. When the changed geometry better represents target-favored labels, the induced rank changes can yield shorter intervals.

Tibshirani et al. (2019) introduce density-ratio weights into the conformal p -value. Subsequent work extends this construction to label shift (Podkopaev and Ramdas, 2021; Plassier et al., 2023) and other shift regimes (Barber et al., 2023). In molecular property prediction, our previous work applies marginal label probability ratio weights to split conformal prediction (Lee et al., 2025). Our recent works study related Bayesian adjustments in the split-conformal setting. Choi (2026) compares post-hoc predictive tilting with in-training posterior adaptation under label shift. Lee et al. (2026) develops split conformal prediction with label-shift-adjusted Bayesian scores. In these split-conformal constructions, the predictive model is fixed before calibration. Target-Aligned Full Conformal Bayes (TA-FCB) instead applies predictive tilting to the candidate-indexed augmented posterior predictive family that arises in Full Conformal Bayes. We summarize our contributions below.

- **Target-aligned Full Conformal Bayes.** We formulate TA-FCB by incorporating label-density-ratio information into both post-ranking aggregation and pre-ranking score construction. The pre-ranking tilt changes the geometry of the candidate-dependent Bayesian scores before rank comparisons are formed, and the resulting changes in the ranks and acceptance sets can produce shorter prediction intervals.

The efficiency improvement therefore follows from the modified score geometry, not from a separate correction objective.

- **Candidate-augmented posterior predictive tilting.** In Full Conformal Bayes, each candidate label induces a different augmented posterior predictive. We define predictive tilting on this candidate-indexed family rather than on a single predictive model fixed before calibration. This candidate-dependent construction distinguishes TA-FCB from related split-conformal approaches and preserves the coupling between candidate augmentation and full-conformal ranking.

We instantiate TA-FCB with Bayesian ridge regression under an exponential label tilt. Candidate augmentation reduces to a rank-one posterior update, and predictive tilting reduces to a variance-scaled shift of the Gaussian predictive mean. The resulting implementation preserves the full-conformal construction without candidate-wise model refitting or importance sampling.

The finite-sample validity statement uses oracle label-density ratios and permutation-equivariant score construction (Tibshirani et al., 2019; Barber et al., 2023). Estimated-ratio variants are evaluated empirically. The perturbation bound is used only to interpret aggregation-stage weight error and is not a finite-sample guarantee for estimated-weight TA-FCB.

2. Related Work

Conformal Bayes and full conformal prediction. Conformal prediction was introduced by Vovk et al. (2005); Shafer and Vovk (2008) and gives finite-sample coverage guarantees under exchangeability without distributional assumptions (Angelopoulos and Bates, 2023). Full conformal prediction (Lei et al., 2018) evaluates the test point symmetrically with the training data via candidate augmentation. This comes at the cost of one model update per candidate label. Recent work explores full conformal prediction for neural regressors via Gauss–Newton influence functions (Tailor et al., 2025), complementary to the closed-form BRR update we use. Fong and Holmes (2021) proposed Conformal Bayes, which uses the Bayesian posterior predictive density as the nonconformity score and handles candidate augmentation via an add-one-in (AOI) importance-sampling identity over posterior samples. Our work starts from exactly this construction and asks what happens to it under label shift. Their AOI identity is a computational device for candidate augmentation, so it leaves both the source-to-target label measure and the geometry induced by label-shift tilting untouched. Recent work studies predictive-level and posterior-level adaptation in split Conformal Bayes (Choi, 2026). Our focus is the full conformal setting, where the predictive and its geometry vary with every candidate label.

Weighted conformal prediction and label shift. Tibshirani et al. (2019) establish the weighted exchangeability framework for conformal prediction under covariate shift, and introduce the weighted p -value (Eq. (5)). Subsequent work extends this to label shift (Podkopaev and Ramdas, 2021; Plassier et al., 2023) and to feedback covariate shift (Fannjiang et al., 2022). All of these approaches reweight already-formed rank comparisons. They are *post-ranking* corrections in the terminology of this paper. Barber et al. (2023) characterize when weighted conformal prediction retains coverage guarantees beyond exchangeability.

Our prior work applies this weighted quantile correction to split conformal prediction under label shift in molecular property prediction (Lee et al., 2025). It discretizes the continuous label when estimating the density ratio. Lee et al. (2026) further studies split conformal prediction with label-shift-adjusted Bayesian scores. In these split procedures, the predictive model is fixed before calibration. The score tilt therefore acts on one fitted predictive. The present work uses a continuous label-density ratio in Full Conformal Bayes. Each candidate label changes the augmented posterior predictive, so pre-ranking tilting is applied candidate-wise. Section 3.3 develops this construction.

Label shift estimation. Label shift estimation is a well-studied problem (Lipton et al., 2018; Azizzadenesheli et al., 2019; Garg et al., 2020; Sugiyama et al., 2012). Existing approaches include moment-matching methods and discriminative density-ratio estimators. For continuous labels, the exponential tilt family gives a tractable parametric model (Azizzadenesheli et al., 2019; Podkopaev and Ramdas, 2021). We use a discriminative estimator in this family. The bias introduced by pseudo-labeling is collected in Appendix A together with an empirical correction. Weight estimation is a practical layer in this paper. The structural claims of Sections 3.2 and 3.3, together with the closed-form BRR instantiation in Section 3.4, are stated under the oracle ratio.

Uncertainty quantification in molecular property prediction. Conformal methods have been adapted to molecular property prediction and drug-target interaction under various shifts (Laghuvarapu et al., 2023; Rakhshaninejad et al., 2025; Antoniuk et al., 2026). Large-scale chemical language representations (Ross et al., 2022; Chithrananda et al., 2020; Irwin et al., 2022) are common backbones for these tasks, and our experiments use one. The method is framed at the level of the Bayesian predictive and the conformal procedure, so it does not depend on a particular representation.

3. Method

We organize the method as follows. Section 3.1 reviews Conformal Bayes and fixes notation. Section 3.2 defines the post-ranking weighted full conformal p -value under continuous label shift. Section 3.3 defines pre-ranking tilting of the candidate-dependent posterior predictive and shows how it changes score geometry before rank formation. Section 3.4 gives a closed-form Bayesian ridge instantiation. Section 3.5 provides an implementation-level comparison with split conformal tilting. The terms pre-ranking and post-ranking indicate where each operation enters the full conformal p -value. The distinctive feature of TA-FCB is the candidate-dependent score geometry generated by full conformal augmentation.

3.1. Conformal Bayes

We review Conformal Bayes (Fong and Holmes, 2021) in the standard exchangeable setting, which also fixes notation for the label-shift analysis that follows. Let $\{(H_i, Y_i)\}_{i=1}^n$ and (H^*, Y^*) be exchangeable random pairs, with $H_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. Let $D_s = \{(h_i, y_i)\}_{i=1}^n$ denote the observed source sample. Let $\mathcal{S}((h, y); D)$ be a nonconformity score that is invariant to permutations of the dataset argument D . For a miscoverage level $\alpha \in (0, 1)$ and

a candidate label $y \in \mathcal{Y}$, the full conformal p -value at h^* is

$$\pi(y; h^*) = \frac{1}{n+1} \left(\sum_{i=1}^n \mathbf{1}\{S_i(y) \geq S^*(y)\} + 1 \right), \quad (1)$$

where $S_i(y) = \mathcal{S}((h_i, y_i); D^{(y)})$ and $S^*(y) = \mathcal{S}((h^*, y); D^{(y)})$. Both scores are computed on the augmented dataset

$$D^{(y)} = D_s \cup \{(h^*, y)\}.$$

The prediction set $C_\alpha(h^*) = \{y : \pi(y; h^*) > \alpha\}$ satisfies $\Pr(Y^* \in C_\alpha(H^*)) \geq 1 - \alpha$ under exchangeability (Vovk et al., 2005). All scores $\{S_i(y)\}_{i=1}^n$ and $S^*(y)$ depend on the candidate y through the augmented data $D^{(y)}$. This candidate dependence is the feature used below to distinguish post-ranking aggregation from rank formation.

Fong and Holmes (2021) instantiate this procedure with a Bayesian nonconformity score. Given a Bayesian model $p(y \mid h, \theta)$ with prior $p(\theta)$, define the candidate-augmented posterior predictive as

$$q^{(y)}(u \mid h) = \int p(u \mid h, \theta) p(\theta \mid D^{(y)}) d\theta.$$

Here the superscript (y) indicates that the posterior is conditioned on the augmented data $D^{(y)}$, while u denotes a generic response value. The Bayesian nonconformity scores are

$$S_i(y) = -\log q^{(y)}(y_i \mid h_i), \quad S^*(y) = -\log q^{(y)}(y \mid h^*). \quad (2)$$

Two features of Eq. (2) are used below. First, the score is candidate-dependent. Split conformal fixes the score after training, whereas in full conformal prediction each candidate label has its own augmented posterior predictive, which gives the candidate-indexed family $\{q^{(y)}\}_{y \in \mathcal{Y}}$. Second, the score is shaped by the posterior predictive learned from the observed source sample. Under label shift, rank comparisons may then be formed under a source-learned predictive geometry that is not aligned with the target domain.

Direct evaluation of Eq. (2) for each candidate y would require recomputing the augmented posterior. Fong and Holmes (2021) avoid this by observing that the augmented posterior is proportional to the base posterior reweighted by the likelihood of the new pair (h^*, y) ,

$$p(\theta \mid D^{(y)}) \propto p(y \mid h^*, \theta) p(\theta \mid D_s), \quad (3)$$

Given posterior samples $\{\theta_m\}_{m=1}^M$ from $p(\theta \mid D_s)$, the augmented posterior predictive can then be computed by self-normalized importance sampling with weights $\omega_m(y) \propto p(y \mid h^*, \theta_m)$. These weights correct posterior computation under candidate augmentation and are unrelated to the source-to-target label-density ratio. AOI therefore makes the candidate-indexed predictive family tractable, while the label-shift construction acts on the predictive geometry and on post-ranking aggregation.

3.2. Post-Ranking Aggregation under Label Shift

Let $p_s(h, y)$ and $p_t(h, y)$ be the source and target joint distributions over representations $h \in \mathbb{R}^d$ and labels $y \in \mathbb{R}$.

Assumption 1 (Label shift):

The source and target joint distributions satisfy

$$p_t(h \mid y) = p_s(h \mid y), \quad p_t(y) \neq p_s(y), \quad (4)$$

Assumption 2 (Absolute continuity):

The target label distribution is absolutely continuous with respect to the source label distribution:

$$p_s(y) > 0 \quad \text{whenever} \quad p_t(y) > 0.$$

Equivalently, the label-density ratio $w(y) = \frac{p_t(y)}{p_s(y)}$ is well defined on the support of p_t .

Assumption 1 and 2 are the structural data-generating settings used throughout this section. Assumption 1 specifies the label-shift setting, in which the conditional distribution from label to representation is shared between domains while the label marginal differs. Assumption 2 ensures that the target label distribution is absolutely continuous with respect to the source distribution, so that the label-density ratio $w(y)$ is well defined on the support of p_t . We observe labeled source data $D_s = \{(h_i, y_i)\}_{i=1}^n$ and target a prediction interval $C_\alpha(h^\star) \subset \mathbb{R}$ for a fresh target point $(H^\star, Y^\star) \sim p_t$ at miscoverage level α . Estimation of w is discussed after the target-aligned construction. Until then we work with the oracle ratio to isolate the structural issues. The finite-sample coverage statements additionally rely on oracle, data-independent label-density ratios and permutation-symmetric score construction. These oracle-validity conditions are stated in Appendix B.

The full conformal rank procedure of Section 3.1 has two stages. In the formation stage, the candidate-dependent scores $\{S_i(y)\}$ and $S^\star(y)$ are computed. In the aggregation stage, these scores are turned into a p -value by the rank sum in Eq. (1). This subsection concerns the aggregation measure. Rank aggregation is source-aligned because each rank comparison in Eq. (1) contributes mass $1/(n+1)$ regardless of the label y_i of the corresponding calibration point. Since the y_i are drawn from p_s , the p -value estimates a probability under the source label marginal. This is harmless in the exchangeable setting. Under label shift, however, the aggregation targets the source label marginal rather than the target label marginal.

The weighted conformal construction of Tibshirani et al. (2019) replaces the uniform mass $1/(n+1)$ in Eq. (1) with normalized weights proportional to $w(y_i)$ and $w(y)$, yielding

$$\pi_w(y; h^\star) = \frac{\sum_{i=1}^n w(y_i) \mathbf{1}\{S_i(y) \geq S^\star(y)\} + w(y)}{\sum_{i=1}^n w(y_i) + w(y)}. \quad (5)$$

Under oracle weights, Assumptions 1 and 2, and the symmetry conditions in Appendix B, Eq. (5) provides target-domain coverage. The scores $\{S_i(y)\}$ and $S^\star(y)$ remain unchanged. This is compatible with validity because the argument depends on weighted exchangeability and score symmetry. W-FCB therefore provides the weighted reference for studying how pre-ranking tilting changes candidate-dependent score geometry.

3.3. Pre-Ranking Candidate-Dependent Predictive Tilting

TA-FCB changes the candidate-dependent score geometry before rank formation. The source posterior predictive varies with the candidate label through posterior augmentation.

To connect the population label-density ratio to the posterior predictive conditioned on the observed source data, we use the following source representativeness assumption.

Assumption 3 (Source representativeness):

The observed source data are representative of the source population in the sense that

$$p_s(y \mid D_s) \approx p_s(y).$$

Under posterior concentration, this condition holds in the limit $n_s \rightarrow \infty$ and is mild for sufficiently large training samples.

Assumption 3 is used only to connect the population-level label-density ratio to the training-data-conditioned predictive distribution. In particular, it justifies applying $w(y) = \frac{p_t(y)}{p_s(y)}$ to a source predictive conditioned on D_s . The post-ranking weighted aggregation does not rely on this approximation. For the candidate-indexed predictive family, we define

$$q_t^{(y)}(u \mid h) = \frac{w(u) q_s^{(y)}(u \mid h)}{Z^{(y)}(h)}, \quad Z^{(y)}(h) = \int w(v) q_s^{(y)}(v \mid h) dv, \quad (6)$$

where the superscript (y) indicates that the predictive is candidate-augmented. It is computed from the posterior under $D^{(y)}$. Taking negative logarithms,

$$-\log q_t^{(y)}(u \mid h) = \underbrace{-\log q_s^{(y)}(u \mid h)}_{\text{base Bayesian score}} \underbrace{-\log w(u)}_{\text{tilt shift}} + \underbrace{\log Z^{(y)}(h)}_{\text{normalizer}}. \quad (7)$$

We define the *target-aligned Bayesian nonconformity score* as the negative log target-aligned augmented posterior predictive,

$$S_i^{\text{TA}}(y) = -\log q_t^{(y)}(y_i \mid h_i), \quad S^{\star, \text{TA}}(y) = -\log q_t^{(y)}(y \mid h^*). \quad (8)$$

These scores remain candidate-dependent, as in the full Conformal Bayes rank procedure. Each score depends on the evaluated label and on the candidate-augmented posterior through $q_t^{(y)}$.

This candidate-wise construction is specific to Full Conformal Bayes. In split conformal prediction, the predictive is fixed after training and tilting produces one score function. In Full Conformal Bayes, each candidate induces a different $q_s^{(y)}$. Applying the tilt to the family $\{q_s^{(y)}\}_y$ preserves the coupling between candidate augmentation and rank formation. In Eq. (7), the base term retains the uncertainty-adaptive Bayesian score. The term $-\log w(u)$ shifts the score according to target label preference. The normalizer $\log Z^{(y)}(h)$ preserves a valid predictive density. The resulting geometric change can alter the rank indicator,

$$\mathbf{1}\{S_i^{\text{TA}}(y) \geq S^{\star, \text{TA}}(y)\} \neq \mathbf{1}\{S_i(y) \geq S^{\star}(y)\}, \quad (9)$$

so tilting can reverse rank comparisons that are sensitive to the source predictive geometry.

The preceding distinction gives two label-shift-aware procedures. *W-FCB* applies the post-ranking aggregation correction alone with the base Bayesian score. *TA-FCB* additionally replaces the base scores with target-aligned scores:

$$\pi_w^w(y; h^*) = \frac{\sum_{i=1}^n w(y_i) \mathbf{1}\{S_i(y) \geq S^{\star}(y)\} + w(y)}{\sum_{i=1}^n w(y_i) + w(y)}, \quad (10)$$

$$\pi_w^{\text{ta}}(y; h^*) = \frac{\sum_{i=1}^n w(y_i) \mathbf{1}\{S_i^{\text{TA}}(y) \geq S^{\star, \text{TA}}(y)\} + w(y)}{\sum_{i=1}^n w(y_i) + w(y)}. \quad (11)$$

The prediction set is $C_\alpha^\bullet(h^*) = \{y : \pi_w^\bullet(y; h^*) > \alpha\}$, with $\bullet \in \{w, \text{ta}\}$. Equations (10) and (11) locate the two operations in the full conformal construction. Post-ranking weighting assigns mass to the rank indicators. Pre-ranking tilting changes the score geometry that generates those indicators. TA-FCB retains the weighted-exchangeability mechanism of W-FCB, and its interval length changes only through the candidate-wise rank changes that the tilted geometry induces. Shorter intervals are therefore an empirical consequence of the geometric change rather than an additional validity mechanism.

When the oracle label-density ratio is used and the score-producing algorithm is permutation equivariant over the augmented dataset, the weighted-exchangeability argument of Tibshirani et al. (2019) applies to both π_w^w and π_w^{ta} . Under these conditions, both oracle prediction sets satisfy $\Pr_t\{Y^* \in C_\alpha^\bullet(H^*)\} \geq 1 - \alpha$. The coverage guarantee follows from oracle weighted exchangeability. Predictive tilting is not an additional requirement for validity. Its role in the proof is to preserve permutation equivariance of the score construction. The computational BRR implementation freezes empirical-Bayes hyperparameters estimated from the source sample and is treated as an approximation to the fully symmetric construction in Appendix B.

For the practical methods, we additionally observe unlabeled target inputs $\{h_j^t\}_{j=1}^m$. We estimate w from source labels and target pseudo-labels via logistic regression. We model the log-odds as linear in y , which corresponds to the exponential tilt family $p_t(y) \propto p_s(y) \exp(\beta_1 y)$ (Azizzadenesheli et al., 2019; Podkopaev and Ramdas, 2021). Estimated weights take the form $\hat{w}(y) = \exp(\hat{\beta}_0 + \hat{\beta}_1 y)$. We use pseudo-labels $\bar{\mu}(h_j^t)$, the predictive means of the source Bayesian model at each target input, as target surrogates. The exponential tilt is a *modeling* choice for the weight estimator rather than a structural assumption. The post-ranking weighting and pre-ranking predictive tilting are defined for a general w . The exponential family is used here for estimation and for the closed-form Bayesian ridge instantiation in Section 3.4. The point pseudo-label estimator produces a target surrogate that is under-dispersed relative to the fitted BRR posterior predictive because it retains only the predictive mean. We examine the resulting effect on density-ratio estimation empirically in Section 4.4 and give the corresponding model-based variance comparison in Appendix A.2. All finite-sample validity claims in this subsection are oracle-weighted. Estimated-weight behavior is reported as an additional practical layer.

3.4. Bayesian Ridge Instantiation

We now instantiate TA-FCB with Bayesian ridge regression (BRR), where candidate augmentation becomes a rank-one posterior update. Under an exponential label tilt, pre-ranking tilting becomes a mean shift of the Gaussian predictive. These two identities make the candidate-dependent score geometry explicit without changing the conformal construction. Let ϕ be a frozen molecular encoder and $h = \phi(x) \in \mathbb{R}^d$. The BRR head is

$$p(y \mid h, \theta) = \mathcal{N}(y \mid \theta^\top h, \lambda^{-1}), \quad p(\theta \mid \alpha_{\text{brr}}) = \mathcal{N}(\theta \mid 0, \alpha_{\text{brr}}^{-1} I). \quad (12)$$

The precision hyperparameters are estimated by empirical Bayes on the observed source sample and are then held fixed during candidate augmentation. Conjugacy yields a Gaussian source posterior over θ with mean $\mu_n = \lambda \Sigma_n H^\top \mathbf{y}$ and covariance $\Sigma_n = (\alpha_{\text{brr}} I + \lambda H^\top H)^{-1}$, where H stacks the source representations and $\mathbf{y}$ the labels. The source posterior predictive

is

$$q_s(y | h) = \mathcal{N}(y | \mu_n^\top h, \sigma^2(h)), \quad \sigma^2(h) = \sigma_{\text{ale}}^2 + h^\top \Sigma_n h, \quad (13)$$

with $\sigma_{\text{ale}}^2 \triangleq \lambda^{-1}$.

For each candidate $y \in \mathcal{Y}_{\text{grid}}$, the full conformal procedure requires the augmented posterior under $D^{(y)} = D_s \cup \{(h^*, y)\}$. Adding one observation gives a rank-one update of the BRR posterior precision. The Sherman–Morrison formula gives the augmented covariance

$$\tilde{\Sigma} = \Sigma_n - \frac{\sigma_h \sigma_h^\top}{\sigma^{*2}}, \quad \sigma_h = \Sigma_n h^*, \quad \sigma^{*2} = \sigma_{\text{ale}}^2 + h^{*\top} \sigma_h, \quad (14)$$

and the augmented predictive variances and mean

$$\tilde{\sigma}_i^2 = \sigma_{\text{ale}}^2 + h_i^\top \tilde{\Sigma} h_i, \quad \tilde{\mu}_n(y) = \lambda \tilde{\Sigma} (H^\top \mathbf{y} + h^* y). \quad (15)$$

Both $\tilde{\Sigma}$ and $\{\tilde{\sigma}_i^2\}_{i=1}^n$ are independent of the candidate y and can be precomputed once per test point. The mean $\tilde{\mu}_n(y)$ varies with y only through the additive term $\lambda \tilde{\Sigma} h^* y$. The per-candidate cost therefore reduces to a vector scaling and addition, with no model retraining. The precision hyperparameters λ and α_{brr} are fixed at their source-data empirical-Bayes estimates throughout candidate evaluation. Conditional on these fixed values, the rank-one update is exact for the BRR posterior mean and covariance. We do not re-estimate them for each candidate-augmented dataset, since that would eliminate the computational advantage of the closed form. The implemented BRR procedure is therefore a source-frozen empirical-Bayes approximation to the fully permutation-equivariant construction considered in Appendix B.

Substituting the augmented Gaussian predictive into the Bayesian score Eq. (2) gives the base calibration score

$$S_i(y) = \frac{(y_i - \tilde{\mu}_n(y)^\top h_i)^2}{2\tilde{\sigma}_i^2} + \frac{1}{2} \log \tilde{\sigma}_i^2 + \text{const}, \quad (16)$$

with the analogous test score

$$S^*(y) = \frac{(y - \tilde{\mu}_n(y)^\top h^*)^2}{2\tilde{\sigma}^{*2}} + \frac{1}{2} \log \tilde{\sigma}^{*2} + \text{const}, \quad \tilde{\sigma}^{*2} = \sigma_{\text{ale}}^2 + h^{*\top} \tilde{\Sigma} h^*.$$

In the calibration scores, the candidate y enters only through the augmented posterior mean $\tilde{\mu}_n(y)$. In the test score, y also appears as the evaluated candidate label.

For the candidate-augmented Gaussian predictive

$$q_s^{(y)}(u | h) = \mathcal{N}(u | \tilde{\mu}_n(y)^\top h, \tilde{\sigma}(h)^2), \quad \tilde{\sigma}(h)^2 = \sigma_{\text{ale}}^2 + h^\top \tilde{\Sigma} h,$$

and the exponential tilt $w(u) = \exp(\beta_0 + \beta_1 u)$, the normalizer in Eq. (6) admits a closed form via the moment-generating function of the Gaussian,

$$\log Z^{(y)}(h) = \beta_0 + \beta_1 \tilde{\mu}_n(y)^\top h + \frac{1}{2} \beta_1^2 \tilde{\sigma}(h)^2. \quad (17)$$

Substituting the candidate-augmented Gaussian predictive and Eq. (17) into the tilt decomposition Eq. (7) and simplifying shows that the target-aligned posterior predictive, as a function of u , is itself Gaussian with the same variance and a shifted mean,

$$q_t^{(y)}(u | h) = \mathcal{N}(u | \tilde{\mu}_n(y)^\top h + \beta_1 \tilde{\sigma}(h)^2, \tilde{\sigma}(h)^2) \quad (18)$$

(derivation in Appendix C). The corresponding target-aligned nonconformity score is

$$S_i^{\text{TA}}(y) = \frac{(y_i - [\tilde{\mu}_n(y)^\top h_i + \beta_1 \tilde{\sigma}_i^2])^2}{2\tilde{\sigma}_i^2} + \frac{1}{2} \log \tilde{\sigma}_i^2 + \text{const}, \quad (19)$$

with the analogous expression for $S^{\star, \text{TA}}(y)$. In this setting, pre-ranking tilting shifts the Gaussian predictive mean by $\beta_1 \tilde{\sigma}(h)^2$. The sign of β_1 determines the direction, and the magnitude of the mean shift is proportional to the predictive variance. The shift therefore changes score level sets and pairwise rank margins before post-ranking aggregation. It does not impose shorter intervals directly. Interval length changes arise when candidates cross the acceptance boundary after the geometry changes.

Algorithm 1 summarizes the full procedure given a weight function and an exponential-tilt slope. In oracle experiments, these are the population ratio w and slope β_1 . In practical runs, they are replaced by estimates $\hat{w}$ and $\hat{\beta}_1$ obtained by the practical weight-estimation procedure described at the end of Section 3.3. Given a test point $h^\star$, the rank-one update (Sherman–Morrison) is computed once, yielding $\tilde{\Sigma}$ and $\{\tilde{\sigma}_i^2\}$ that are shared across all candidates. For each grid point y , the augmented mean follows the affine form $\tilde{\mu}_n(y) = a + by$, and the projected means $h_i^\top \tilde{\mu}_n(y)$ can be evaluated from precomputed affine coefficients. The weighted rank sum then produces the p -value. After these precomputations, the candidate scan costs $O(nK)$ and requires neither retraining nor importance sampling.

The BRR closed form separates computation from geometry: the rank-one update evaluates candidate-dependent posterior predictives efficiently, and the Gaussian exponential tilt expresses the geometric change as a mean shift. Neither identity changes the definition of TA-FCB.

The candidate-augmented posterior predictive can be approximated through the AOI identity of Fong and Holmes (2021) in Eq. (3). Given posterior samples $\{\theta_m\}_{m=1}^M$ from $p(\theta \mid D_s)$, each candidate predictive can be evaluated by self-normalized importance sampling with $\omega_m(y) \propto p(y \mid h^\star, \theta_m)$. The BRR instantiation avoids this candidate-specific sampling step and provides a closed-form expression for the score geometry.

3.5. Relation to Split Conformal Tilting

Recent split-conformal formulations adjust a fixed Bayesian predictive before weighted calibration (Choi, 2026; Lee et al., 2026). In split conformal, BRR is fitted once on D_{train} . The resulting source predictive is tilted once and then held fixed. Calibration scores are computed on D_{cal} , and a weighted split conformal quantile is used for test-time acceptance. The training data are not augmented with any test candidate. The split conformal method therefore performs neither candidate augmentation nor a candidate-dependent posterior update.

TA-FCB uses the merged labeled sample D_s and retains full conformal candidate augmentation. Each candidate label y defines

$$q_s^{(y)}(u \mid h) = \int p(u \mid h, \theta) p(\theta \mid D^{(y)}) d\theta. \quad (20)$$

Algorithm 1: Label-Shift-Aware Full Conformal Bayes with BRR (W-FCB / TA-FCB)

Input: $D_s = \{(h_i, y_i)\}_{i=1}^n$; test point h^* ; miscoverage α ; weight function w ; tilt slope β_1 ; grid $\mathcal{Y}_{\text{grid}}$; mode $\in \{\text{base}, \text{TA}\}$

Output: finite-grid prediction interval $\hat{C}_\alpha^\bullet(h^*)$ with the one-cell endpoint expansion

Fit BRR on D_s : obtain μ_n, Σ_n, λ ;

$\sigma_h \leftarrow \Sigma_n h^*, \sigma^{*2} \leftarrow \sigma_{\text{ale}}^2 + h^{*\top} \sigma_h, \tilde{\Sigma} \leftarrow \Sigma_n - \sigma_h \sigma_h^\top / \sigma^{*2}$; // rank-1 update

(Sherman-Morrison), Eq. (14)

$\tilde{\sigma}_i^2 \leftarrow \sigma_{\text{ale}}^2 + h_i^\top \tilde{\Sigma} h_i$ for $i = 1, \dots, n$; $\tilde{\sigma}^{*2} \leftarrow \sigma_{\text{ale}}^2 + h^{*\top} \tilde{\Sigma} h^*$;

Precompute $a \leftarrow \lambda \tilde{\Sigma} H^\top \mathbf{y}$ and $b \leftarrow \lambda \tilde{\Sigma} h^*$ so that $\tilde{\mu}_n(y) = a + by$;

Precompute $c_i \leftarrow h_i^\top a$ and $d_i \leftarrow h_i^\top b$ for $i = 1, \dots, n$, and $c^* \leftarrow h^{*\top} a, d^* \leftarrow h^{*\top} b$;

Precompute $W \leftarrow \sum_{i=1}^n w(y_i)$;

for $y \in \mathcal{Y}_{\text{grid}}$ **do**

$\eta_i(y) \leftarrow c_i + d_i y$ for $i = 1, \dots, n$; $\eta^*(y) \leftarrow c^* + d^* y$;

if mode = TA **then**

$S_i(y) \leftarrow -\log \mathcal{N}(y_i \mid \eta_i(y) + \beta_1 \tilde{\sigma}_i^2, \tilde{\sigma}_i^2)$ for $i = 1, \dots, n$;

$S^*(y) \leftarrow -\log \mathcal{N}(y \mid \eta^*(y) + \beta_1 \tilde{\sigma}^{*2}, \tilde{\sigma}^{*2})$;

else

$S_i(y) \leftarrow -\log \mathcal{N}(y_i \mid \eta_i(y), \tilde{\sigma}_i^2)$ for $i = 1, \dots, n$;

$S^*(y) \leftarrow -\log \mathcal{N}(y \mid \eta^*(y), \tilde{\sigma}^{*2})$;

end

$\pi_w^\bullet(y) \leftarrow [\sum_i w(y_i) \mathbf{1}\{S_i(y) \geq S^*(y)\} + w(y)] / [W + w(y)]$;

end

$J \leftarrow \{k \in \{1, \dots, K\} : \pi_w^\bullet(y_k; h^*) > \alpha\}$;

$j_L \leftarrow \min J; j_U \leftarrow \max J$;

return $[y_{\max(1, j_L-1)}, y_{\min(K, j_U+1)}]$;

Hence the target-aligned predictive is defined candidate-wise as

$$q_t^{(y)}(u \mid h) = \frac{w(u) q_s^{(y)}(u \mid h)}{\int w(v) q_s^{(y)}(v \mid h) dv}, \quad \forall y \in \mathcal{Y}. \quad (21)$$

For the BRR implementation, the Sherman–Morrison covariance update is computed once per test input. The candidate-dependent augmented mean is then evaluated in the affine form $\tilde{\mu}_n(y) = a + by$ over the candidate grid. Every candidate produces its own tilted calibration scores and test score. These scores enter the full conformal p -value together with the candidate weight $w(y)$.

The split conformal method tilts one fitted predictive geometry and calibrates it on a held-out set. TA-FCB tilts the candidate-indexed geometry created by full conformal augmentation and aggregates candidate-coupled rank indicators. The empirical comparison in Section 4.5 is end-to-end because the methods also use different labeled-data allocations.

4. Experiments

The experiments evaluate post-ranking weighting and the geometric effect of pre-ranking predictive tilting. Full conformal Bayes (FCB) provides the unadjusted reference. Weighted full conformal Bayes (W-FCB) fixes the weighted aggregation while retaining the base Bayesian score. TA-FCB uses the same weighted aggregation after changing the candidate-dependent score geometry. The main results use oracle weights so that the comparison between W-FCB and TA-FCB is not confounded by label-density-ratio estimation.

4.1. Experimental Setup

We evaluate the proposed corrections on two molecular property prediction tasks from the Therapeutics Data Commons (Huang et al., 2021). The tasks are AqSolDB for aqueous solubility, reported as $\log S$ with $n = 9,982$ (Sorkun et al., 2019), and Lipophilicity for $\log D$ with $n = 4,200$ (Wu et al., 2018; Wenlock and Tomkinson, 2015). We induce label shift at test time by exponential tilting of the source label marginal, with $p_t(y) \propto p_s(y) \exp(\beta_1 y)$. The shift is controlled by a single scalar β_1 . We sweep β_1 over $\{0, -0.1, \dots, -0.5\}$, where $\beta_1 = 0$ corresponds to the no-shift setting. For split conformal methods, we partition the data into $D_{\text{train}}/D_{\text{cal}}/D_{\text{weight}}/D_{\text{test}}$ with proportions 0.4/0.2/0.2/0.2. For full conformal methods, we merge D_{train} and D_{cal} , which gives a 0.6/0.2/0.2 split over training, weight-estimation, and test sets. Both methods use the same D_{weight} split when label-shift weights are estimated. Molecular representations h are extracted using a SMILES-pretrained BART encoder (Ross et al., 2022). The encoder is kept frozen. We fit the Bayesian ridge regression head on the resulting 768-dimensional representations by empirical Bayes. For numerical implementation, we evaluate the candidate-wise p -value on $K = 100$ equally spaced labels over the finite range $[\mu_n^\top h^* - 4\tilde{\sigma}^*, \mu_n^\top h^* + 4\tilde{\sigma}^*]$. We report the interval $[y_{j_L-1}, y_{j_U+1}]$, where j_L and j_U are the first and last accepted grid indices and the endpoints are expanded by one grid cell. Under the in-range regularity conditions stated in Appendix D, this expansion reduces boundary discretization error for the grid-detected acceptance component. The same candidate range, grid resolution, and endpoint rule are used for all full-conformal methods. As this implementation evaluates only a finite range of candidates, its coverage properties are assessed empirically. The finite-sample result established in Appendix B applies to the unrestricted oracle construction. All experiments are averaged over 1000 random seeds, with the target coverage level set to $1 - \alpha = 0.9$.

We compare four methods built from the same BRR model class and the same oracle ratio. **FCB** denotes Full Conformal Bayes without label-shift adjustment. It uses Eq. (1) with the base Bayesian score. **W-FCB** uses the post-ranking weighted p -value in Eq. (10). **TA-FCB** uses the same aggregation after replacing the base score with the pre-ranking tilted score in Eq. (11). The three full-conformal methods fit BRR on the merged $D_{\text{train}} \cup D_{\text{cal}}$ sample. **TA-SCB** fits BRR once on D_{train} , computes fixed tilted scores on D_{cal} , and applies a weighted split conformal quantile. TA-SCB does not use candidate augmentation or the rank-one full-conformal update. It is included as an end-to-end split conformal counterpart. For the dataset-size experiment, TA-SCB subsamples D_{train} and D_{cal} separately while preserving their original split ratio, 2 : 1. In contrast, the full-conformal methods train on the union of the subsampled D_{train} and D_{cal} splits. Throughout this experiment, the sizes of D_{weight} and D_{test} are kept fixed across all subsampling levels.

As a practical analysis separate from the main results, we also report results based on estimated label-shift weights. The weights are given by $\hat{w}(y) = \exp(\hat{\beta}_0 + \hat{\beta}_1 y)$ and are fit by logistic regression on source labels and target pseudo-labels. These results do not carry the finite-sample validity guarantee of the oracle-weighted construction. We include them as an empirical check of practical implementability.

4.2. Main Results

All results in this subsection use oracle label-shift weights. We discuss estimated-weight results later in Section 4.4. FCB, W-FCB, and TA-FCB share the same fitted BRR model and labeled-data allocation. Their differences are therefore attributable to the full conformal construction. TA-SCB uses the same BRR model class and oracle ratio but retains a separate calibration split. Comparisons involving TA-SCB are interpreted as end-to-end full-versus-split comparisons.

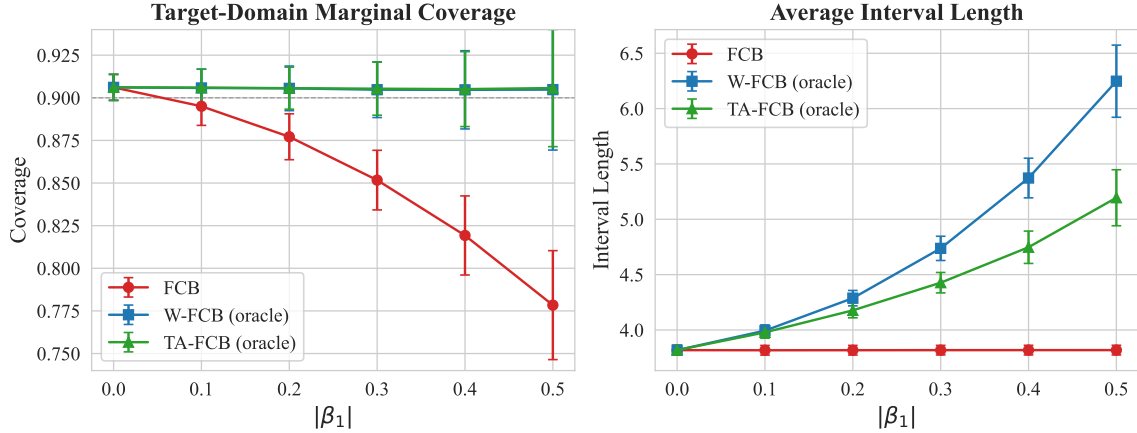


Figure 1: Target-domain marginal coverage (left) and average interval length (right) under increasing label shift $|\beta_1|$ on solubility with BRR. FCB denotes unweighted full Conformal Bayes; W-FCB (oracle) is the post-ranking correction with oracle weights; TA-FCB (oracle) adds the pre-ranking correction. Error bars show ± 1 standard deviation over 1000 seeds.

Figure 1 reports target-domain marginal coverage and average interval length on the solubility task as a function of $|\beta_1|$. The corresponding numerical values are reported in Table 2 in Appendix E. FCB, the unweighted baseline, degrades sharply under label shift. Its coverage decreases monotonically as $|\beta_1|$ grows, dropping from 0.906 at $\beta_1 = 0$ to 0.778 at $|\beta_1| = 0.5$ on the solubility task. In FCB, both rank stages remain source-aligned. Rank comparisons are therefore neither formed nor aggregated under the target label distribution. This behavior is the expected consequence of applying Conformal Bayes without label-shift correction. Oracle W-FCB reaches near-nominal coverage at all shift magnitudes, including 0.905 at $|\beta_1| = 0.5$ (Table 3 in Appendix E). This pattern is consistent with the theory of Tibshirani et al. (2019); Barber et al. (2023). Introducing weights into rank aggregation reduces the effective sample size used to estimate the calibration quantile (Tibshirani et al.,

2019), causing W-FCB to produce longer intervals than the unweighted FCB baseline. W-FCB therefore serves as the post-ranking reference for isolating the geometric effect of pre-ranking tilting.

Oracle TA-FCB remains near 0.9 coverage and produces shorter intervals than oracle W-FCB at every shift magnitude (Table 3 in Appendix E). This coverage behavior is consistent with the oracle validity result in Appendix B. The reported values use the common finite-grid implementation. At $|\beta_1| = 0.5$, the average interval length is 5.195 for TA-FCB and 6.248 for W-FCB. The reduction is approximately 17%, while coverage remains essentially unchanged. The two methods use the same fitted BRR model, oracle ratio, and post-ranking weighted p -value. Their difference is the geometry of the candidate-dependent score. The variance-scaled mean shift changes pairwise rank margins and moves some candidates across the acceptance boundary. The shorter interval is the observed consequence of these rank changes.

Results for the lipophilicity task are reported in Appendix F and Table 4. The qualitative pattern is consistent across the two tasks. FCB undercovers on lipophilicity, with coverage decreasing to 0.859 at $|\beta_1| = 0.5$. At matched coverage, oracle TA-FCB reduces average interval length by 6.6% relative to oracle W-FCB, from 3.004 to 2.807 according to Table 4 in Appendix F. The smaller relative gain on lipophilicity is expected. The label range of lipophilicity is narrower than that of solubility. As a result, the same value of $|\beta_1|$ induces a weaker change in the predictive geometry, leaving less room for predictive tilting to shorten the intervals.

In Section 4.4, we show that the same pre-ranking correction does not require oracle access. The target-aligned surrogate (TAS) procedure is derived from the same tilting identity and recovers near-oracle coverage across all shift magnitudes (Figure 3). The structural and practical results above isolate the two corrections under matched conformal constructions. A complementary question is whether the full conformal rank procedure provides an efficiency advantage over its split conformal counterpart when the pre-ranking correction is matched. We examine this question separately in Section 4.5.

4.3. Diagnostics for the Pre-Ranking Correction

Section 3.3 defines predictive tilting as a change in candidate-dependent score geometry through the rank indicators $\mathbf{1}\{S_i^{\text{TA}}(y) \geq S^{\star, \text{TA}}(y)\}$. We examine this mechanism by comparing $\pi_w^{\text{w}}(\cdot; h^\star)$ and $\pi_w^{\text{ta}}(\cdot; h^\star)$ with identical oracle weights for each $|\beta_1|$. Any difference between the two p -value curves then comes from rank flips induced by the tilted score. A change in curve shape can move the accepted-set boundaries. The diagnostic below measures these rank flips and the resulting endpoint movement over the candidate grid.

To quantify how tilting reshapes the p -value curve, we use the decomposition of the p -value difference derived in Appendix G:

$$\pi_w^{\text{ta}}(y; h^\star) - \pi_w^{\text{w}}(y; h^\star) = \frac{\sum_{i=1}^n w_i [R_i^{\text{ta}}(y) - R_i^{\text{w}}(y)]}{\sum_{i=1}^n w_i + w(y)}, \quad (22)$$

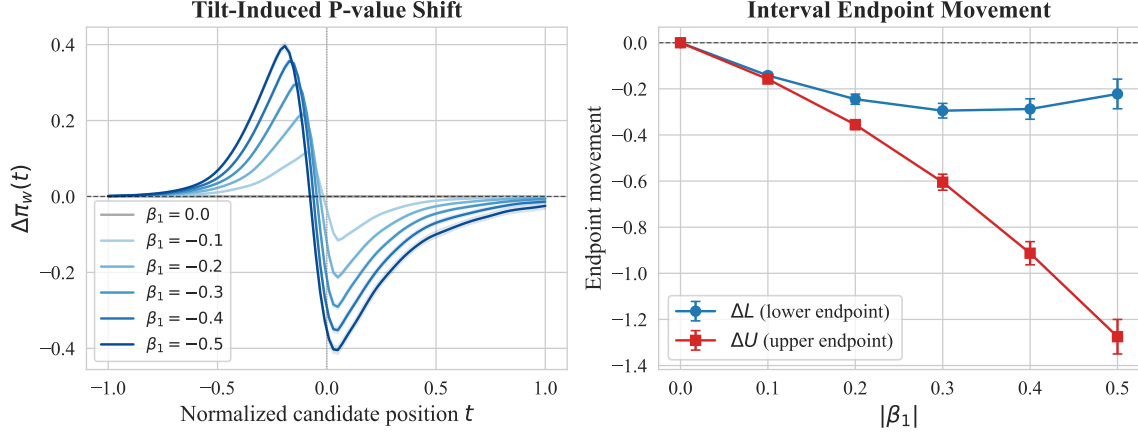


Figure 2: Rank-shaping diagnostics on solubility (1000 seeds). **Left:** p -value difference $\pi_w^{\text{ta}}(y; h^*) - \pi_w^{\text{w}}(y; h^*)$ averaged over all test points and seeds, plotted against normalized candidate position $t = (y - \mu_n^\top h^*) / (4\tilde{\sigma}^*)$. Positive values indicate that target-aligned scoring raises the p -value relative to source scoring; negative values indicate a reduction. Under $\beta_1 < 0$, negative mass concentrates in the upper tail ($t > 0$), growing monotonically with $|\beta_1|$. **Right:** Mean endpoint movement ΔL (lower) and ΔU (upper) as a function of $|\beta_1|$, averaged over 1000 seeds (error bars: ± 1 std.). The upper endpoint contracts substantially more than the lower, consistent with the asymmetric p -value difference pattern in the left panel.

where $R_i^{\text{ta}}(y) = \mathbf{1}\{S_i^{\text{TA}}(y) \geq S^{*,\text{TA}}(y)\}$ and $R_i^{\text{w}}(y) = \mathbf{1}\{S_i(y) \geq S^*(y)\}$. Positive values indicate that tilting raises the p -value, whereas negative values indicate that it lowers the p -value.

Figure 2 reports two summaries of this effect. The left panel shows the p -value difference $\pi_w^{\text{ta}}(y; h^*) - \pi_w^{\text{w}}(y; h^*)$ averaged over all test points and 1000 random seeds as a function of the normalized candidate position $t = (y - \mu_n^\top h^*) / (4\tilde{\sigma}^*)$. The right panel shows the induced movements of the lower and upper endpoints, denoted by ΔL and ΔU . Under $\hat{\beta}_1 < 0$, the p -value difference is negative in the upper tail ($t > 0$). This directly removes upper-tail candidates from the prediction set. Near the lower boundary, the difference is positive and partially offsets the leftward movement of the interval. The resulting asymmetry gives $|\Delta U| > |\Delta L|$ at every value of $|\beta_1|$. Moreover, $|\Delta U|$ grows approximately linearly with $|\beta_1|$ and reaches 1.275 at $\beta_1 = -0.5$. In contrast, $|\Delta L|$ peaks near 0.29 at $|\beta_1| = 0.3$ and decreases thereafter.

4.4. Practical Feasibility

The structural results above use oracle label-shift weights. In practice, the label-shift weights are unknown and have to be estimated. Figure 3 compares three weight-estimation strategies under TA-FCB. Point pseudo-labeling retains only the BRR predictive mean. The resulting target surrogate is under-dispersed relative to the fitted posterior predictive, and

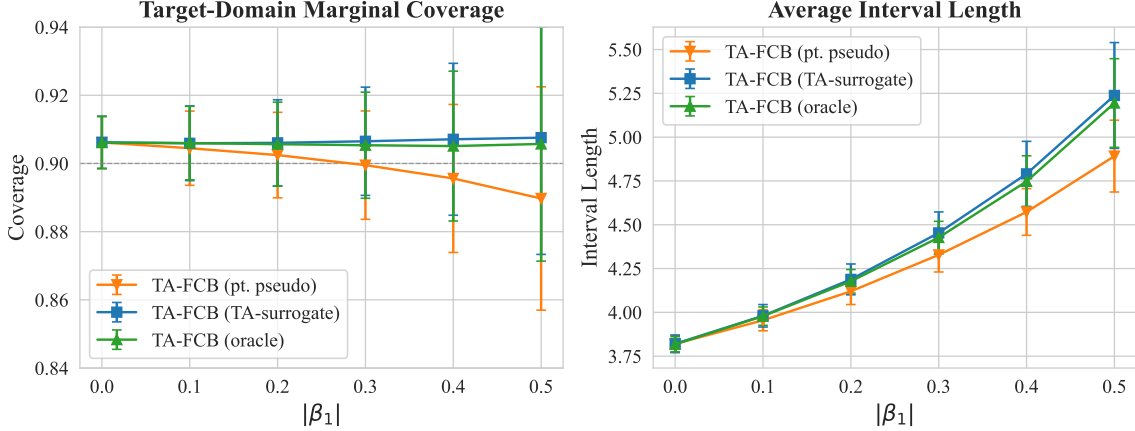


Figure 3: Practical feasibility of the pre-ranking correction on solubility (1000 seeds). TA-FCB (pt. pseudo) uses point pseudo-labels for weight estimation and exhibits growing undercoverage with $|\beta_1|$ (pseudo-label bias). TA-FCB (TA-surrogate) uses target-aligned surrogate generation (Appendix A.3) and recovers near-oracle coverage across all shift magnitudes, which shows that the pre-ranking correction does not require oracle access.

the estimated tilt slope is attenuated in our runs. Target-aligned surrogate generation (TAS) reuses the same tilting identity to shift the surrogate center toward the target domain and reintroduce model-based predictive variation (Appendix A.3). Against the oracle-weight reference, pseudo-label weighting undercovers by more and more as $|\beta_1|$ grows, reaching 0.890 at $|\beta_1| = 0.5$. TAS recovers near-oracle coverage across all shift magnitudes, with coverage of 0.908 compared with 0.906 for the oracle reference. Its interval length remains within 1% of the oracle value, with 5.238 for TAS compared with 5.195 for the oracle. The pre-ranking correction can therefore be implemented with estimated ratios, and in these experiments it performs close to the oracle reference. TAS generates its surrogates from the same posterior-predictive tilting identity that defines the correction.

4.5. Full vs. Split Conformal

Section 4.2 compares W-FCB and TA-FCB within one full-conformal construction. We next compare TA-FCB with TA-SCB. Both use oracle weights and the same Gaussian tilt, but their score construction and labeled-data allocation differ. We report results across shift magnitudes and across dataset sizes.

Figure 4 (top row) reports oracle TA-FCB and oracle TA-SCB on the solubility task across $|\beta_1| \in \{0.0, \dots, 0.5\}$. Both procedures use the same BRR model class, oracle label-shift ratio, and Gaussian predictive tilt. TA-FCB fits BRR on the entire training dataset and evaluates candidate-dependent tilted scores inside the full conformal p -value. TA-SCB fits BRR on the proper training split and calibrates a fixed tilted score on the held-out calibration split. At $|\beta_1| = 0.5$, the average interval length is 5.195 for TA-FCB and 5.432 for TA-SCB, a reduction of about 4.4%, while coverage differs by 0.002. Because score

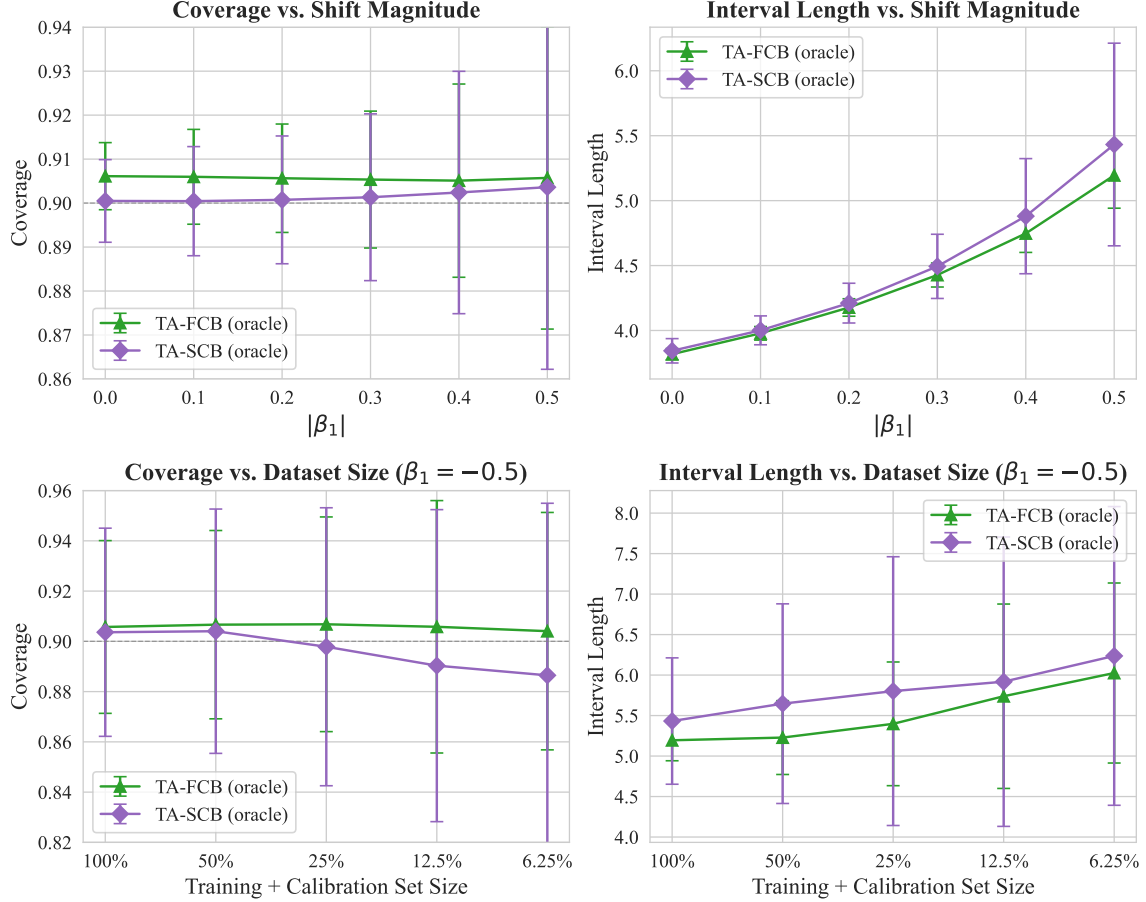


Figure 4: Full and split conformal results on solubility with oracle weights (1000 seeds). The top row reports coverage and interval length across shift magnitudes $|\beta_1|$. The bottom row reports the same metrics under dataset subsampling at fixed $\beta_1 = -0.5$. TA-FCB uses candidate-dependent full-conformal scores on the merged labeled split. TA-SCB uses fixed tilted scores and a held-out calibration split. The comparison is end-to-end.

construction and data allocation both differ, this result is an end-to-end comparison rather than an isolated post-ranking effect. Table 7 in Appendix H reports the numerical values.

Figure 4 (bottom row) reports results on the solubility task at fixed $\beta_1 = -0.5$, with the training set subsampled to $\{100\%, 50\%, 25\%, 12.5\%, 6.25\%\}$ of its original size. The corresponding numerical values are reported in Table 6 in Appendix H. At every subsampling level, the TA-FCB maintains near-nominal coverage while producing shorter interval length. The coverage gap between TA-FCB and TA-SCB widens as data become scarce. At the 6.25% subsampling level, TA-FCB achieves coverage of 0.904, compared with 0.887 for TA-SCB.

This coverage deficit traces to the test-point weight used in the weighted split conformal quantile. Under label shift the weight that weighted exchangeability requires at the test point is $w(Y^*)$, a function of the unobserved test label, so our implementation substitutes the plug-in $w(\bar{\mu}(h^*))$. In Appendix H we recompute the same quantile using each held-out test point’s realized label y^* in place of the plug-in and confirm that coverage is restored to or above nominal at every subsampling level, which isolates the plug-in as the source of the deficit. This is a diagnostic, not an implementable correction: Y^* is exactly what conformal prediction must cover without observing, so substituting its realized value is only possible after the fact. The diagnostic is also not free of cost, as it sharply increases the frequency of unbounded intervals, particularly at small calibration-set sizes; we report these effects in Appendix H. TA-FCB is not exposed to this trade-off, because its candidate-wise construction evaluates the weight $w(y)$ at every candidate y on the grid, so whichever candidate matches the realized label y^* already carries the correct weight without any substitution.

5. Conclusion

We introduced TA-FCB for continuous label shift. In this setting, weighted rank aggregation and candidate-wise predictive tilting serve different roles in Full Conformal Bayes. Post-ranking weighting defines the target-weighted aggregation of rank indicators. The distinctive step is pre-ranking tilting of the candidate-augmented posterior predictive family. This tilt changes Bayesian score geometry and can alter candidate-wise rank comparisons. Under oracle weights and the symmetry conditions in Appendix B, the validity mechanism remains weighted exchangeability. The resulting improvement in interval efficiency is attributable to this geometric change.

The construction gives a target-aligned Bayesian nonconformity score that is well defined within the full conformal rank procedure. Bayesian ridge regression under the exponential tilt family makes both mechanisms explicit in closed form: candidate augmentation reduces to a rank-one posterior update, and pre-ranking tilting reduces to a variance-scaled shift of the Gaussian predictive mean. The implementation is therefore candidate-wise and needs no model refitting.

On two molecular property benchmarks, W-FCB recovers the coverage lost by unweighted FCB. TA-FCB produces shorter intervals at comparable coverage in the reported experiments. The interval changes are explained by rank flips induced by the tilted score geometry. The comparison with TA-SCB is end-to-end because the full and split procedures use different score constructions and labeled-data allocations. Estimated-ratio variants approach oracle performance in the reported experiments but do not carry finite-sample guarantees.

Several limitations remain. First, our Bayesian ridge regression instantiation exploits the Gaussian exponential-tilt pairing to make both corrections explicit and separately measurable. Extending the pre-ranking correction to more general Bayesian models requires combining the candidate-dependent tilt with the AOI identity of Fong and Holmes (2021). This extension requires candidate-specific importance sampling over posterior samples, substantially increasing the computational cost relative to the closed-form BRR implementation. The statistical construction remains unchanged, but the closed-form separation is lost.

We leave a full empirical study of this variant to future work. Second, our validity results cover only oracle label-shift weights. Tighter theoretical control of pseudo-label bias and of the score perturbation induced by estimation error in the pre-ranking correction remains open. Third, we study label shift within the exponential-tilt family. Richer shift structures, including localized shifts, joint label and covariate shifts, and time-varying shifts, lie outside that family and remain open. Finally, the efficiency gain from pre-ranking tilting depends on the Bayesian model being reasonably well specified. Coverage is still guaranteed under oracle label-shift weights and the oracle-validity conditions of Appendix B, because the weighted conformal argument applies to any permutation-symmetric score. The interval-length improvement, however, relies on the tilted predictive distribution approximating the true target posterior predictive distribution. Under severe model misspecification, TA-FCB is therefore expected to retain coverage under oracle weighting, while its interval-length advantage over W-FCB may shrink, disappear, or even reverse.

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Appendix A. Weight Estimation and Pseudo-Label Bias

This appendix collects the practical weight-estimation machinery used in Section 4: the discriminative estimator, the pseudo-label bias, the target-aligned surrogate (TAS) construction that corrects it, and a coverage perturbation bound for the aggregation stage. None of this belongs to the structural scope of the paper (Sections 3.2–3.4), and all of it is stated empirically rather than as part of the validity argument.

A.1. Discriminative Estimator in the Exponential Tilt Family

Under the label-shift assumption Eq. (4), the label-density ratio can be estimated by binary classification between source and target labels (Tibshirani et al., 2019; Lipton et al., 2018; Azizzadenesheli et al., 2019; Podkopaev and Ramdas, 2021). Introducing a domain indicator $D \in \{0, 1\}$,

$$w(y) = \frac{p_t(y)}{p_s(y)} = \frac{p(D=1 | y)}{p(D=0 | y)} \cdot \frac{p(D=0)}{p(D=1)}. \quad (23)$$

In our experimental implementation, the source and target classes used for weight estimation are constructed from D_{weight} and D_{test} , respectively, using equal sample sizes. Therefore, the domain-classification sample has balanced class priors, $p(D=0) = p(D=1) = 1/2$, and the domain-prior correction factor equals one. Modeling logit $p(D = 1 | y) = \beta_0 + \beta_1 y$ with logistic regression yields estimates of the form $\hat{w}(y) = \exp(\hat{\beta}_0 + \hat{\beta}_1 y)$, which lies in the exponential tilt family.

A.2. Variance Compression under Point Pseudo-Labels

We use target pseudo-labels $\tilde{y}_j = \bar{\mu}(h_j^t) = \mu_n^\top h_j^t$, the BRR predictive mean at each target input, as surrogates for the target side of the classifier. Let $\tilde{Y}_{\text{pred}}$ denote the random label represented by the fitted source BRR posterior predictive at a target input:

$$\tilde{Y}_{\text{pred}} | H = h \sim \mathcal{N}(\bar{\mu}(h), \sigma^2(h)), \quad H \sim p_t(h).$$

Conditioning on the fitted BRR model, the law of total variance gives

$$\begin{aligned} \text{Var}_t(\tilde{Y}_{\text{pred}}) &= \text{Var}_t\{\bar{\mu}(H)\} + \mathbb{E}_t\{\sigma^2(H)\}, \\ \text{Var}_t(\tilde{Y}_{\text{point}}) &= \text{Var}_t\{\bar{\mu}(H)\}. \end{aligned} \quad (24)$$

Here, the expectation and variance are taken over target inputs and, for $\tilde{Y}_{\text{pred}}$, over the posterior-predictive draw. Therefore, the point-pseudo-label surrogate has smaller variance than the corresponding BRR predictive surrogate by $\mathbb{E}_t\{\sigma^2(H)\}$.

A.3. Target-Aligned Surrogate Generation

Point pseudo-labeling collapses each fitted BRR posterior predictive to its mean and omits the stochastic component described in Appendix A.2. We therefore construct stochastic target surrogates from a target-aligned predictive approximation rather than using the point mean alone. The sampling term reintroduces model-based predictive variation, while the tilt-induced shift adjusts the predictive center toward the target domain. We call this target-aligned surrogate (TAS) generation.

Construction. Because the exponential-tilt parameter β_1 is unknown, we construct a low-cost initial tilt from the observed shift in the model-predictable component. A first-order expansion of the exponentially tilted expectation around β_1 gives

$$\mathbb{E}_t[\bar{\mu}(H)] - \mathbb{E}_s[\bar{\mu}(H)] = \beta_1 \text{Cov}_s\{\bar{\mu}(H), Y\} + O(\beta_1^2).$$

When $\bar{\mu}(H)$ is a reasonable approximation to the source conditional mean, $\text{Cov}_s\{\bar{\mu}(H), Y\} \approx \text{Var}_s\{\bar{\mu}(H)\}$. This motivates the initialization

$$\hat{\beta}_{\text{MM}} = \frac{\bar{\mu}_t - \bar{\mu}_s}{\hat{\sigma}_s^2}. \quad (25)$$

where

$$\bar{\mu}_t = \frac{1}{|D_{\text{test}}|} \sum_{j \in D_{\text{test}}} \bar{\mu}(h_j^t), \quad \bar{\mu}_s = \frac{1}{|D_{\text{weight}}|} \sum_{i \in D_{\text{weight}}} \bar{\mu}(h_i), \quad \hat{\sigma}_s = \frac{1}{|D_{\text{weight}}| - 1} \sum_{i \in D_{\text{weight}}} (\bar{\mu}(h_i) - \bar{\mu}_s)^2.$$

Eq. (25) is used only to initialize target-aligned surrogate generation; the final ratio slope is subsequently re-estimated by logistic regression on source labels and generated target surrogates. For a Gaussian source posterior predictive, the tilting identity yields a target-aligned posterior predictive with shifted mean and preserved variance; evaluated at $\hat{\beta}_{\text{MM}}$, we draw target surrogate labels as

$$\tilde{y}_j^{\text{TAS}} = \bar{\mu}(h_j^t) + \hat{\beta}_{\text{MM}} \sigma^2(h_j^t) + \sigma(h_j^t) \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, 1). \quad (26)$$

Fitting the source-vs.-surrogate logistic classifier on $(D_{\text{weight}}, \{\tilde{y}_j^{\text{TAS}}\}_{j=1}^m)$ yields the final estimated ratio $\hat{w}(y) = \exp(\hat{\beta}_0 + \hat{\beta}_1 y)$, used in both calibration weighting and the tilted score. TAS therefore uses the same heteroscedastic predictive variance $\sigma^2(h)$ that appears in the posterior-predictive tilting construction. The target-directed shift $\hat{\beta}_{\text{MM}} \sigma^2(h)$ adjusts the surrogate center toward the target domain, while the sampling term reintroduces the stochastic variation represented by the fitted BRR posterior predictive. These operations define a model-based surrogate construction and do not imply exact recovery of the unknown target-label distribution.

Empirical comparison. Table 1 reports coverage and interval length on solubility for pseudo-label, TAS, and oracle weights across shift magnitudes, with matched pre-ranking correction (TA-FCB throughout). Pseudo-label weighting exhibits the residual undercoverage reported in Section 4.4, which grows with $|\beta_1|$. TAS recovers near-nominal coverage at every shift magnitude, tracking oracle within 0.002. The interval length is within 1% of the oracle length. Weight-estimation noise therefore does not materially disturb the structural length reduction that the pre-ranking correction provides.

A.4. Coverage Perturbation Bound

The following result is an interpretive tool for the residual coverage gap under estimated weights. It is scoped to the base, non-tilted weighted procedure, in which the nonconformity-score construction does not depend on the weight estimate. Extending the argument to the target-aligned case, where the estimated tilt parameter also enters the scores, is an open direction noted in Section 5.

Table 1: Target-Aligned Surrogate (TAS) versus pseudo-label and oracle weights on solubility, under TA-FCB (post- and pre-ranking). TAS tracks oracle coverage to within 0.002 across all shift magnitudes, while pseudo-label weights undercover. Values are averaged over 1000 seeds.

$ \beta_1 $	Coverage $\uparrow$			Interval Length $\downarrow$		
	pt. pseudo	TA-surrogate	oracle	pt. pseudo	TA-surrogate	oracle
0.0	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	3.820 $\pm$ 0.049	3.821 $\pm$ 0.050	3.818 $\pm$ 0.043
0.1	0.905 $\pm$ 0.011	0.906 $\pm$ 0.011	0.906 $\pm$ 0.011	3.955 $\pm$ 0.060	3.980 $\pm$ 0.064	3.978 $\pm$ 0.052
0.2	0.903 $\pm$ 0.013	0.906 $\pm$ 0.013	0.906 $\pm$ 0.012	4.120 $\pm$ 0.075	4.188 $\pm$ 0.088	4.177 $\pm$ 0.067
0.3	0.900 $\pm$ 0.016	0.907 $\pm$ 0.016	0.905 $\pm$ 0.016	4.328 $\pm$ 0.098	4.453 $\pm$ 0.120	4.427 $\pm$ 0.092
0.4	0.896 $\pm$ 0.022	0.907 $\pm$ 0.022	0.905 $\pm$ 0.022	4.573 $\pm$ 0.133	4.790 $\pm$ 0.186	4.747 $\pm$ 0.146
0.5	0.890 $\pm$ 0.033	0.908 $\pm$ 0.034	0.906 $\pm$ 0.034	4.892 $\pm$ 0.205	5.238 $\pm$ 0.302	5.195 $\pm$ 0.253

Assumption 4 (Linear log-density ratio):

The log-density ratio is linear in y : $\log w(y) = \beta_0 + \beta_1 y$ for some $\beta_1 \in \mathbb{R}$. Estimated weights are also in the family $\hat{w}(y) = \exp(\hat{\beta}_0 + \hat{\beta}_1 y)$, so $\hat{w}(y)/w(y) = \exp(\delta_0 + \delta_1 y)$ with $\delta_1 = \hat{\beta}_1 - \beta_1$.

Proposition 1 (Coverage perturbation under multiplicative weight error) Under Assumption 4, suppose the scores $S_1, \dots, S_n, S^*$ are fixed (do not depend on $\hat{w}$), the oracle W-FCB procedure satisfies $\Pr_t(Y^* \in C_{\alpha'}) \geq 1 - \alpha'$ for all $\alpha' \in (0, 1)$, and the label set $\mathcal{Y}$ has diameter R_y . Let $\hat{C}_\alpha$ be the prediction set produced by W-FCB with estimated weights $\hat{w}$ at level α . Then, provided $|\delta_1|R_y < \log(1/\alpha)$,

$$\Pr_t(Y^* \in \hat{C}_\alpha) \geq 1 - \alpha e^{|\delta_1|R_y} = (1 - \alpha) - \alpha(e^{|\delta_1|R_y} - 1). \quad (27)$$

Proof Let $c = e^{-|\delta_1|R_y}$. Since $\hat{w}(y)/w(y) = \exp(\delta_0 + \delta_1 y)$ is monotone in y , the ratio between its maximum and minimum over $\mathcal{Y}$ is $e^{|\delta_1|R_y}$. Then

$$\hat{\pi}_w(y) = \frac{\sum_i \hat{w}(y_i) \mathbf{1}\{S_i \geq S^*\} + \hat{w}(y)}{\sum_i \hat{w}(y_i) + \hat{w}(y)} \geq c \pi_w(y).$$

Hence $\{y : \pi_w(y) > \alpha/c\} \subseteq \{y : \hat{\pi}_w(y) > \alpha\}$, so $C_{\alpha/c} \subseteq \hat{C}_\alpha$, and $\Pr_t(Y^* \in \hat{C}_\alpha) \geq \Pr_t(Y^* \in C_{\alpha/c}) \geq 1 - \alpha/c$. $\blacksquare$

Heuristic interpretation. Point pseudo-labeling retains only the predictive mean and therefore omits the stochastic component represented by the fitted BRR posterior predictive. This omission can distort the source-versus-target surrogate classification problem. The resulting slope error depends on the predictive variance, model misspecification, shift magnitude, and the overlap between the source and surrogate label distributions. We therefore treat the observed attenuation of $\hat{\beta}_1$ as an empirical phenomenon rather than assigning it a universal closed-form scaling law.

Scope. Proposition 1 assumes that the nonconformity scores do not depend on the weight estimate. In the target-aligned variant Eq. (19), $\hat{\beta}_1$ enters both the p -value weights and the scores, so the proposition should be read as a lower bound on the coverage degradation due to weight estimation in the aggregation stage alone. Quantifying the additional coverage effect of score perturbations under estimated weights is left to future work.

Appendix B. Finite-Sample Validity under Oracle Weights

This appendix establishes finite-sample target-domain coverage for W-FCB and TA-FCB under oracle label-density ratios and the stated symmetry conditions. In both procedures, coverage follows from weighted exchangeability. Since TA-FCB uses a different score from W-FCB, the one extra proof step is to verify that predictive tilting preserves permutation equivariance.

Conditions. Besides Assumptions 1–3, the oracle-validity proof additionally requires the following conditions.

- (C1) **Oracle weights:** Both the weighting and tilting use the true label-density ratio, treated as a known population quantity.
- (C2) **Algorithm symmetry:** For each candidate y , the score vector computed from the augmented dataset is permutation-equivariant in its $n + 1$ observations. This property must be preserved by every data-dependent component used to construct the scores, including hyperparameter selection. It holds when hyperparameters are fixed independently of the conformal sample or are re-estimated symmetrically on every candidate-augmented dataset.

Standard Bayesian ridge regression with fixed, data-independent hyperparameters satisfies (C2), as does empirical-Bayes BRR when the hyperparameters are re-estimated symmetrically on every augmented dataset. The source-frozen empirical-Bayes implementation in Section 3.4 is treated as an approximation to this symmetric construction.

Weighted exchangeability. To establish finite-sample coverage, we evaluate the conformal procedure at the realized target label Y^* . Let $Z_i = (H_i, Y_i)$ for $i = 1, \dots, n$, and let $Z_{n+1} = (H^*, Y^*)$. Under Assumption 1,

$$p_t(h, y) = p_t(y)p_s(h \mid y) = w(y)p_s(y)p_s(h \mid y) = w(y)p_s(h, y).$$

Hence the ordered joint density satisfies

$$f(z_{1:n+1}) = \prod_{i=1}^n p_s(z_i) p_t(z_{n+1}) = \left(\prod_{i=1}^{n+1} p_s(z_i) \right) w(y_{n+1}).$$

Writing

$$g(z_{1:n+1}) = \prod_{i=1}^{n+1} p_s(z_i),$$

we have

$$f(z_{1:n+1}) = g(z_{1:n+1}) w(y_{n+1}),$$

where g is permutation-invariant. Therefore the augmented sample satisfies the weighted-exchangeability condition of Tibshirani et al. (2019). Conditional on the unordered augmented sample,

$$\Pr(J = j \mid \{Z_1, \dots, Z_{n+1}\}) = \frac{w(Y_j)}{\sum_{k=1}^{n+1} w(Y_k)},$$

which is exactly the normalized weighting used in Eqs. 10–11.

Score equivariance is preserved by tilting. For an augmented dataset $D = (Z_1, \dots, Z_{n+1})$, let $\mathbf{S}^s(D) = (S_1^s(D), \dots, S_{n+1}^s(D))$ denote the source score vector.

Condition (C2) states that the score construction is permutation-equivariant: for every permutation ρ , $\mathbf{S}^s(\rho D) = \rho \mathbf{S}^s(D)$.

The target-aligned predictive is

$$q_t(u \mid h) = \frac{w(u) q_s(u \mid h)}{\int w(v) q_s(v \mid h) dv}.$$

Because the oracle weight function $w(\cdot)$ is a fixed population quantity under (C1), independent of the observed data, the same permutation-equivariance property is preserved after tilting:

$$\mathbf{S}^{\text{TA}}(\rho D) = \rho \mathbf{S}^{\text{TA}}(D).$$

Therefore, predictive tilting changes the score values, but not the permutation structure required by the weighted-exchangeability argument.

Validity. Under Assumptions 1 and 2, together with (C1)–(C2), the weighted exchangeability theorem of Tibshirani et al. (2019) applies verbatim to both π_w^w and π_w^{ta} . Following the argument, for any $\alpha \in (0, 1)$ the unrestricted oracle prediction set $C_\alpha^\bullet(h^\star) = \{y : \pi_w^\bullet(y; h^\star) > \alpha\}$ satisfies

$$\Pr_{(H^\star, Y^\star) \sim p_t} \{Y^\star \in C_\alpha^\bullet(H^\star)\} \geq 1 - \alpha, \quad (28)$$

where $\bullet \in \{w, \text{ta}\}$. Thus, exact finite-sample target-domain coverage holds for the oracle constructions satisfying (C1) and (C2). The source-frozen empirical-Bayes implementation in Section 3.4 and the finite-range grid implementation in Appendix D are treated as practical approximations to this symmetric oracle construction. The same argument does not extend to estimated weights. With an estimated ratio $\hat{w}$, the finite-sample validity proof for oracle weighted conformal prediction no longer applies directly. For this reason, the estimated-weight results of Section 4 and Appendix A are stated empirically and not as finite-sample guarantees.

Appendix C. Target-Aligned Predictive under Gaussian Tilt

This appendix derives the closed-form expression used in Section 3.4.

Starting from the tilting identity Eq. (6),

$$-\log q_t^{(y)}(u \mid h) = -\log q_s^{(y)}(u \mid h) - \log w(u) + \log Z^{(y)}(h), \quad Z^{(y)}(h) = \int w(u) q_s^{(y)}(u \mid h) du. \quad ((7) \text{ revisited})$$

Specializing to BRR under exponential tilt, let $q_s^{(y)}(u | h) = \mathcal{N}(u | m, s^2)$ with $m = \tilde{\mu}_n(y)^\top h$ and $s^2 = \tilde{\sigma}(h)^2$, and $w(u) = \exp(\beta_0 + \beta_1 u)$. The normalizer $Z^{(y)}(h)$ is the moment-generating function of the Gaussian evaluated at β_1 :

$$Z^{(y)}(h) = \int e^{\beta_0 + \beta_1 u} \mathcal{N}(u | m, s^2) du = \exp(\beta_0 + \beta_1 m + \frac{1}{2} \beta_1^2 s^2). \quad (29)$$

Substituting into Eq. (7) and collecting Gaussian terms,

$$\begin{aligned} -\log q_t^{(y)}(u | h) &= \frac{(u - m)^2}{2s^2} + \frac{1}{2} \log(2\pi s^2) - (\beta_0 + \beta_1 u) + \beta_0 + \beta_1 m + \frac{1}{2} \beta_1^2 s^2 \\ &= \frac{(u - (m + \beta_1 s^2))^2}{2s^2} + \frac{1}{2} \log(2\pi s^2), \end{aligned} \quad (30)$$

where the last equality follows by completing the square. Therefore $q_t^{(y)}(u | h) = \mathcal{N}(u | m + \beta_1 s^2, s^2)$, which is Eq. (18).

The target-aligned Bayesian score in Eq. (19) is then simply the negative log Gaussian density with the shifted mean. The predictive variance $\tilde{\sigma}(h)^2$ is unchanged, so tilting preserves the original Gaussian uncertainty penalty, including both the standardized residual term and the log-variance term. Consequently, the uncertainty adaptivity of the score is preserved.

Appendix D. Finite-Grid Implementation and Endpoint Expansion

In the experiments, the candidate-wise p -value is evaluated on a finite grid $\mathcal{Y}_{\text{grid}} = \{y_1, \dots, y_K\}$ contained in a test-input-dependent numerical range. Let

$$C_{\alpha, \text{range}}^\bullet = \{y \in [y_1, y_K] : \pi_w(y; h^\star) > \alpha\}, \quad \bullet \in \{\text{w}, \text{ta}\}$$

denote the acceptance set restricted to this evaluated range. The grid procedure approximates this in-range set and does not evaluate candidates outside $[y_1, y_K]$.

Endpoint expansion within the evaluated range. Let j_L and j_U denote the first and last grid indices satisfying $\pi_w^\bullet(y_j; h^\star) > \alpha$. We return the interval $\hat{C}_\alpha^\bullet = [y_{j_L-1}, y_{j_U+1}]$ with the convention $y_{j_L-1} = y_1$ if $j_L = 1$ and $y_{j_U+1} = y_K$ if $j_U = K$.

Suppose that $C_{\alpha, \text{range}}^\bullet$ is a single interval, contains at least one accepted grid point, and adjacent evaluated grid points bracket each of its two boundaries. Under these in-range conditions, the one-cell expansion contains the grid-detected acceptance component. Specifically, if j_L and j_U are the first and last accepted grid indices, then the left and right boundaries lie no farther than one grid cell beyond y_{j_L} and y_{j_U} , respectively. Hence the endpoint expansion controls boundary discretization error within the evaluated range. The relation between the selected numerical range and the unrestricted oracle acceptance set is outside this in-range discretization argument. Accordingly, we treat $\hat{C}_\alpha^\bullet$ as the implemented finite-grid procedure and assess its target-domain coverage empirically.

Relation to prior work. [Chen et al. \(2018\)](#) formalize the grid-discretization problem for conformal prediction and propose rounded-data methods. Our construction is simpler. It requires only the superset argument above and does not modify the score or the data.

The cost is a small length increase of approximately two grid cells. Because the same grid and endpoint rule are applied to all full-conformal methods, the reported comparisons use a common discretization protocol.

Connectedness for BRR. For Bayesian ridge regression with Gaussian predictive, the acceptance set is empirically always connected in our experiments. We monitor the disconnected-interval rate (the fraction of test points for which the grid-level acceptance set $\{y_k : \pi_w^\bullet(y_k; h^*) > \alpha\}$ is not contiguous) across all seeds and shift magnitudes. It is exactly zero in every solubility and lipophilicity trial. Thus, no disconnected component was detected at the resolution of the evaluated grids.

Empirical verification. On solubility at $\beta_1 = 0$ (no shift, target coverage 0.9) with $K = 100$ grid points and 1000 seeds, conservative endpoints yield coverage 0.906 ± 0.008 (above nominal, as expected from the superset construction) with interval length 3.818 ± 0.043 . The one-cell expansion adds approximately $2\Delta \approx 0.1$ to the reported interval length.

Appendix E. Oracle versus Pseudo-Label Weights

Table 2 reports the main structural results (FCB, W-FCB (oracle), TA-FCB (oracle)) across shift magnitudes on solubility. Table 3 additionally compares pseudo-label and oracle weights for W-FCB and TA-FCB. The coverage gap under pseudo-label weights is the residual undercoverage analyzed in Appendix A.2.

Table 2: Solubility results under BRR (1000 seeds). FCB (unweighted baseline), oracle W-FCB (post-ranking correction), and oracle TA-FCB (post- and pre-ranking corrections).

$ \beta_1 $	Coverage $\uparrow$			Interval Length $\downarrow$		
	FCB	W-FCB (oracle)	TA-FCB (oracle)	FCB	W-FCB (oracle)	TA-FCB (oracle)
0.0	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	3.818 $\pm$ 0.043	3.818 $\pm$ 0.043	3.818 $\pm$ 0.043
0.1	0.895 $\pm$ 0.011	0.906 $\pm$ 0.011	0.906 $\pm$ 0.011	3.817 $\pm$ 0.044	3.994 $\pm$ 0.052	3.978 $\pm$ 0.052
0.2	0.877 $\pm$ 0.014	0.906 $\pm$ 0.013	0.906 $\pm$ 0.012	3.817 $\pm$ 0.044	4.287 $\pm$ 0.070	4.177 $\pm$ 0.067
0.3	0.852 $\pm$ 0.018	0.905 $\pm$ 0.016	0.905 $\pm$ 0.016	3.818 $\pm$ 0.043	4.737 $\pm$ 0.110	4.427 $\pm$ 0.092
0.4	0.819 $\pm$ 0.023	0.905 $\pm$ 0.023	0.905 $\pm$ 0.022	3.818 $\pm$ 0.044	5.373 $\pm$ 0.179	4.747 $\pm$ 0.146
0.5	0.778 $\pm$ 0.032	0.905 $\pm$ 0.036	0.906 $\pm$ 0.034	3.818 $\pm$ 0.044	6.248 $\pm$ 0.326	5.195 $\pm$ 0.253

Table 3: Pseudo-label vs. oracle weights for W-FCB and TA-FCB on solubility.

β_1	Coverage $\uparrow$				Interval Length $\downarrow$			
	W-FCB		TA-FCB		W-FCB		TA-FCB	
	pseudo	oracle	pseudo	oracle	pseudo	oracle	pseudo	oracle
0.0	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	0.906 $\pm$ 0.008	3.820 $\pm$ 0.047	3.818 $\pm$ 0.043	3.820 $\pm$ 0.049	3.818 $\pm$ 0.043
-0.1	0.904 $\pm$ 0.011	0.906 $\pm$ 0.011	0.904 $\pm$ 0.011	0.906 $\pm$ 0.011	3.964 $\pm$ 0.064	3.994 $\pm$ 0.052	3.955 $\pm$ 0.060	3.978 $\pm$ 0.052
-0.2	0.901 $\pm$ 0.013	0.906 $\pm$ 0.013	0.902 $\pm$ 0.013	0.906 $\pm$ 0.012	4.198 $\pm$ 0.089	4.287 $\pm$ 0.070	4.120 $\pm$ 0.075	4.177 $\pm$ 0.067
-0.3	0.896 $\pm$ 0.017	0.905 $\pm$ 0.016	0.900 $\pm$ 0.016	0.905 $\pm$ 0.016	4.550 $\pm$ 0.133	4.737 $\pm$ 0.110	4.328 $\pm$ 0.098	4.427 $\pm$ 0.092
-0.4	0.891 $\pm$ 0.023	0.905 $\pm$ 0.023	0.896 $\pm$ 0.022	0.905 $\pm$ 0.022	5.037 $\pm$ 0.195	5.373 $\pm$ 0.179	4.573 $\pm$ 0.133	4.747 $\pm$ 0.146
-0.5	0.884 $\pm$ 0.034	0.905 $\pm$ 0.035	0.890 $\pm$ 0.033	0.906 $\pm$ 0.034	5.650 $\pm$ 0.293	6.248 $\pm$ 0.326	4.892 $\pm$ 0.205	5.195 $\pm$ 0.253

Appendix F. Lipophilicity Results

We report results on lipophilicity under the same protocol as solubility. The qualitative pattern matches the solubility results of Section 4.2. Unweighted FCB undercovers under shift. Oracle W-FCB restores coverage. Oracle TA-FCB shortens intervals at matched coverage, and TA-FCB produces shorter intervals than TA-SCB.

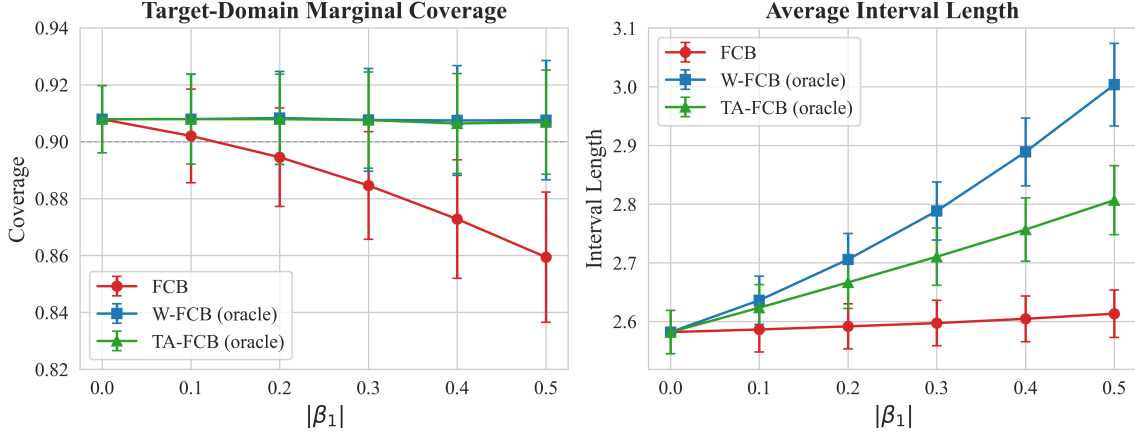


Figure 5: Lipophilicity results under BRR with increasing $|\beta_1|$.

Table 4: BRR conformal prediction on lipophilicity. Structural view matches Table 2: FCB (unweighted baseline), oracle W-FCB (post-ranking correction), and oracle TA-FCB (post- and pre-ranking corrections). Split-conformal counterpart is reported in Section 4.5.

β_1	Coverage $\uparrow$			Interval Length $\downarrow$		
	FCB	W-FCB (oracle)	TA-FCB (oracle)	FCB	W-FCB (oracle)	TA-FCB (oracle)
0.0	0.908 $\pm$ 0.012	0.908 $\pm$ 0.012	0.908 $\pm$ 0.012	2.582 $\pm$ 0.037	2.582 $\pm$ 0.037	2.582 $\pm$ 0.037
-0.1	0.902 $\pm$ 0.017	0.908 $\pm$ 0.016	0.908 $\pm$ 0.016	2.586 $\pm$ 0.038	2.636 $\pm$ 0.041	2.623 $\pm$ 0.039
-0.2	0.895 $\pm$ 0.017	0.908 $\pm$ 0.016	0.908 $\pm$ 0.016	2.592 $\pm$ 0.038	2.706 $\pm$ 0.044	2.667 $\pm$ 0.044
-0.3	0.885 $\pm$ 0.019	0.908 $\pm$ 0.018	0.908 $\pm$ 0.017	2.597 $\pm$ 0.039	2.788 $\pm$ 0.050	2.710 $\pm$ 0.049
-0.4	0.873 $\pm$ 0.021	0.908 $\pm$ 0.019	0.906 $\pm$ 0.018	2.604 $\pm$ 0.039	2.889 $\pm$ 0.058	2.757 $\pm$ 0.054
-0.5	0.859 $\pm$ 0.023	0.908 $\pm$ 0.021	0.907 $\pm$ 0.018	2.613 $\pm$ 0.040	3.004 $\pm$ 0.070	2.807 $\pm$ 0.059

Appendix G. Rank-Flip Decomposition of the p -Value Difference

This appendix derives the exact decomposition used in Section 4.3 and provides the score-level identity that explains the asymmetric endpoint movement observed in Figure 2.

Exact p -value decomposition. Fix a test input h^* and a candidate y . Let $W = \sum_{i=1}^n w(y_i)$ and write the two weighted p -values as

$$\pi_w^{\text{ta}}(y; h^*) = \frac{R_t(y) + w(y)}{W + w(y)}, \quad \pi_w^{\text{w}}(y; h^*) = \frac{R_s(y) + w(y)}{W + w(y)},$$

where $R_t(y) = \sum_i w_i R_i^t(y)$, $R_s(y) = \sum_i w_i R_i^s(y)$, and $R_i^t(y) = \mathbf{1}\{S_i^{\text{TA}}(y) \geq S^{*,\text{TA}}(y)\}$, $R_i^s(y) = \mathbf{1}\{S_i(y) \geq S^*(y)\}$. Since both p -values share the same denominator,

$$\pi_w^{\text{ta}}(y; h^*) - \pi_w^{\text{w}}(y; h^*) = \frac{F_{\text{net}}(y)}{W + w(y)}, \quad F_{\text{net}}(y) = \sum_{i=1}^n w_i [R_i^t(y) - R_i^s(y)]. \quad (31)$$

This is an exact identity, not an approximation. $F_{\text{net}}(y)$ is the *signed weighted rank-flip mass*. Positive flips ($R_i^s = 0 \rightarrow R_i^t = 1$) raise the p -value. Negative flips ($R_i^s = 1 \rightarrow R_i^t = 0$) lower it. Target-aligned scoring removes a candidate from the accepted set if and only if $F_{\text{net}}(y) < 0$ is large enough to push $\pi_t(y)$ below α .

Score-level mechanism under Gaussian BRR. Define the pairwise score margins $D_i^s(y) = S_i(y) - S^*(y)$ and $D_i^t(y) = S_i^{\text{ta}}(y) - S^{*,\text{ta}}(y)$. Using the BRR target-aligned score Eq. (19) and the base score Eq. (16), and writing $r_i(y) = y_i - \tilde{\mu}_n(y)^\top h_i$ and $r_*(y) = y - \tilde{\mu}_n(y)^\top h^*$ for the augmented residuals, a direct calculation gives

$$D_i^t(y) - D_i^s(y) = -\beta_1 (r_i(y) - r_*(y)) + \frac{1}{2} \beta_1^2 (\tilde{\sigma}_i^2 - \tilde{\sigma}^{*2}). \quad (32)$$

This identity is exact and holds pointwise for every calibration index i and every candidate y .

Mechanism for upper-tail asymmetry. For $\beta_1 < 0$ and $r_i(y) < r_*(y)$, the first-order term $-\beta_1 (r_i(y) - r_*(y))$ is negative. When this term dominates the variance-dependent second-order term, the tilted margin is smaller than the base margin, making a negative rank flip possible:

$$0 \leq D_i^s(y) < \beta_1 (r_i(y) - r_*(y)) - \frac{1}{2} \beta_1^2 (\tilde{\sigma}_i^2 - \tilde{\sigma}^{*2}). \quad (33)$$

Near the lower boundary, $r_*(y) < 0$ and calibration points with $r_i(y) > r_*(y)$ receive a positive first-order shift in $D_i^t - D_i^s$, making positive flips ($R_i = 0 \rightarrow R_i^{\text{ta}} = 1$) possible instead. This sign reversal between the upper and lower tails is the score-level origin of the asymmetric p -value difference pattern in Figure 2 (left).

We do not claim that $|\Delta U| > |\Delta L|$ holds unconditionally. The actual dominance of negative flips near the upper boundary is finite-sample dependent and is verified empirically by the signed rank-flip diagnostic in Figure 2.

Numerical values. Table 5 reports the mean and standard deviation of ΔL and ΔU across 1000 seeds on solubility for each shift magnitude.

Appendix H. Calibration-Set Size Sensitivity

Table 6 and Table 7 report the numerical values corresponding to Figure 4 in the main text.

Table 5: Interval endpoint movement ΔL (lower) and ΔU (upper) under target-aligned scoring on solubility (1000 seeds, mean $\pm$ s.d.). Values are the difference between TA-FCB and W-FCB endpoints; negative values indicate leftward movement.

$ \beta_1 $	ΔL	ΔU
0.0	0.000 ± 0.000	0.000 ± 0.000
0.1	-0.142 ± 0.014	-0.158 ± 0.014
0.2	-0.245 ± 0.022	-0.355 ± 0.021
0.3	-0.295 ± 0.032	-0.605 ± 0.035
0.4	-0.288 ± 0.044	-0.913 ± 0.050
0.5	-0.222 ± 0.064	-1.275 ± 0.075

Table 6: Oracle TA-FCB and oracle TA-SCB on solubility at fixed $\beta_1 = -0.5$ across dataset sizes (1000 seeds). TA-FCB uses entire training data and candidate-dependent scores. TA-SCB uses a held-out calibration set and fixed tilted scores. The last column counts the seeds in which TA-SCB returns an unbounded interval; the reported TA-SCB lengths are averages over the remaining seeds. TA-FCB returns no unbounded interval at any set size.

Set size	Coverage $\uparrow$		Interval Length $\downarrow$		n_∞
	TA-FCB (oracle)	TA-SCB (oracle)	TA-FCB (oracle)	TA-SCB (oracle)	
100%	0.906 ± 0.034	0.904 ± 0.041	5.195 ± 0.253	5.432 ± 0.780	0
50%	0.907 ± 0.038	0.904 ± 0.049	5.228 ± 0.456	5.647 ± 1.233	0
25%	0.907 ± 0.043	0.898 ± 0.055	5.398 ± 0.764	5.802 ± 1.660	2
12.5%	0.906 ± 0.050	0.890 ± 0.062	5.739 ± 1.139	5.918 ± 1.787	47
6.25%	0.904 ± 0.047	0.887 ± 0.069	6.025 ± 1.112	6.236 ± 1.844	304

Diagnostic: substituting the realized test label for the plug-in weight. Table 6 reports TA-SCB with the practical plug-in $w(\bar{\mu}(h^*))$ in place of $w(Y^*)$, the weight that weighted exchangeability requires at the test point. To confirm that this plug-in is the source of the coverage deficit, we recompute the same weighted split-conformal quantile using each held-out test point’s realized label y^* in place of the plug-in, that is with test weight $w(y^*)$. Table 8 reports the result. Coverage is restored to at or above nominal at every subsampling level, which isolates the plug-in substitution as the cause of the deficit reported in the main text. This is a diagnostic, not an implementable correction: Y^* is exactly the quantity conformal prediction must cover without observing, so substituting its realized value is only possible after the fact, once the held-out label is already known.

The diagnostic is not free of cost. Because $w(u) = e^{\beta_1 u}$ is convex, $w(Y^*)$ exceeds the plug-in $w(\bar{\mu}(h^*))$ in expectation, so using the realized label pushes more test points’ weight past the threshold needed for a finite weighted quantile, $w(y^*) > \frac{\alpha}{(1-\alpha)} \sum_i w(y_i)$, and the resulting interval is unbounded. The seed count in Table 8 reports, for each subsampling level, how many of the 1000 seeds contain at least one test point with an unbounded interval;

Table 7: Oracle TA-FCB versus oracle TA-SCB on solubility across shift magnitudes over 1000 seeds. Both use the same BRR model class, oracle ratio, and Gaussian tilt. Their score construction and labeled-data allocation differ.

$ \beta_1 $	Coverage $\uparrow$		Interval Length $\downarrow$	
	TA-SCB (oracle)	TA-FCB (oracle)	TA-SCB (oracle)	TA-FCB (oracle)
0.0	0.901 $\pm$ 0.009	0.906 $\pm$ 0.008	3.844 $\pm$ 0.094	3.818 $\pm$ 0.043
0.1	0.900 $\pm$ 0.012	0.906 $\pm$ 0.011	4.001 $\pm$ 0.111	3.978 $\pm$ 0.052
0.2	0.901 $\pm$ 0.015	0.906 $\pm$ 0.012	4.211 $\pm$ 0.153	4.177 $\pm$ 0.067
0.3	0.901 $\pm$ 0.019	0.905 $\pm$ 0.016	4.494 $\pm$ 0.247	4.427 $\pm$ 0.092
0.4	0.902 $\pm$ 0.028	0.905 $\pm$ 0.022	4.880 $\pm$ 0.443	4.747 $\pm$ 0.146
0.5	0.904 $\pm$ 0.041	0.906 $\pm$ 0.034	5.432 $\pm$ 0.780	5.195 $\pm$ 0.253

the reported length averages only the seeds without one. At 6.25% subsampling, 989 of 1000 seeds contain an unbounded interval, so the reported length there is an average over the remaining 11 seeds and should be read with that caveat.

Table 8: Diagnostic: TA-SCB with the test-point weight evaluated at each held-out test point’s realized label y^* instead of the plug-in $w(\bar{\mu}(h^*))$, at $\beta_1 = -0.5$ across dataset sizes (1000 seeds). This diagnostic is not implementable as a conformal procedure because it requires the unobserved test label, and is reported solely to isolate the source of the coverage deficit in Table 6. Here, n_∞ denotes the number of seeds containing at least one unbounded interval, and interval length is averaged over the remaining seeds only.

Set size	Coverage $\uparrow$	Interval Length $\downarrow$	n_∞
100%	0.907 $\pm$ 0.042	5.470 $\pm$ 0.780	0
50%	0.912 $\pm$ 0.049	5.734 $\pm$ 1.234	0
25%	0.923 $\pm$ 0.046	6.016 $\pm$ 1.736	377
12.5%	0.928 $\pm$ 0.047	6.621 $\pm$ 2.145	827
6.25%	0.935 $\pm$ 0.048	7.199 $\pm$ 1.992	989